FROM COMPLEX ANALYSIS TO THE GROUP STRUCTURE ON ELLIPTIC CURVE

JINGHUA XI

Abstract. This note aims to derive the group structure on elliptic curves using elementary tools such as Liouville's theorem and argument principle.

1. Lattice and Elliptic functions

Let $w_1, w_2 \in \mathbb{C}$ such that $\Im(w_1/w_2) > 0$. We define the *lattice* associated with w_1, w_2 as $L \coloneqq w_1\mathbb{Z} \oplus w_2\mathbb{Z} \subset \mathbb{C}$.

Associated to this lattice L, we define a *complex torus* as the quotient space $\mathbb{C}/L$. This space inherits the natural structure of a Riemann surface. Within this framework, a meromorphic function on the complex torus can be naturally identified with a meromorphic function on $\mathbb{C}$ possessing a double period. These are termed elliptic functions.

Specifically, an elliptic function associated with lattice L is defined as a meromorphic function f on $\mathbb{C}$ satisfying $f(x) = f(y)$ for any $x, y \in \mathbb{C}$ such that $x - y \in L$. We denote the set of all such functions by $\mathcal{E}_L$.

Remark 1.1. The set $\mathcal{E}_L$ forms a field closed under differentiation.

We introduce some notations initially. Let Π represent the parallelogram in $\mathbb{C}$, defined as $\{(sw_1, tw_2) : s, t \in [0, 1]\}$. For a given $\alpha \in \mathbb{C}$, consider a function f from the set $\mathcal{E}_L$, which exhibits neither zeros nor poles on the boundary $\partial(\alpha+\Pi)$, denoted as $\alpha+\Pi = \{\alpha+z : z \in \Pi\}$. The existence of such α is guaranteed, as zeros and poles in Π are discrete and thus finite, given Π's compactness.

Lemma 1.2. *For any $f \in \mathcal{E}_L$ that is holomorphic on $\partial(\alpha + \Pi)$, we have $\int_{\partial(\alpha+\Pi)} f = 0$.*

Proof. The integral $\int_{\partial(\alpha+\Pi)} f$ can be decomposed into the sum of integrals over each side of the parallelogram:

$$\int_{\alpha}^{\alpha+w_1} f + \int_{\alpha+w_1}^{\alpha+w_1+w_2} f + \int_{\alpha+w_1+w_2}^{\alpha+w_2} f + \int_{\alpha+w_2}^{\alpha} f.$$

Since f is elliptic, by change of variable

$$\int_{\alpha+w_1}^{\alpha+w_1+w_2} f(z)dz = \int_{\alpha}^{\alpha+w_2} f(z + w_1)d(z + w_1) = \int_{\alpha}^{\alpha+w_2} f(z)dz.$$

Likewise,

$$\int_{\alpha+w_2}^{\alpha+w_1+w_2} f(z)dz = \int_{\alpha}^{\alpha+w_1} f(z)dz.$$

This implies $\int_{\partial(\alpha+\Pi)} f = 0$. □

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Proposition 1.3. *For any elliptic function $f \in \mathcal{E}_L$, there are equal numbers of zeros and poles, counted with multiplicities, within the parallelogram Π.*

Proof. It suffices to verifies this in $\alpha + \Pi$, since f is elliptic. Given f has neither zeros nor poles on $\partial(\alpha + \Pi)$, the Argument Principle implies

$$\#Z(f) - \#P(f) = \int_{\partial(\alpha+\Pi)} \frac{f'}{f}$$

, where $\#Z(f), \#P(f)$ denote zeros and poles of f up to multiplicities in Π respectively. Notice $\frac{f'}{f} \in \mathcal{E}_L$ thus it follows that

$$\#Z(f) - \#P(f) = \int_{\partial(\alpha+\Pi)} \frac{f'}{f} = 0$$

by Lemma1.2. □

Proposition 1.4. *For every elliptic function $f \in \mathcal{E}_L$, the sum $\sum_{z\in\alpha+\Pi} ord_z(f) \cdot z$ is an element of the lattice L, where $ord_z(f)$ represents the order of f at the point z.*

Proof. By the generalized argument principle, we have

$$\sum_{z\in\alpha+\Pi} \mathrm{ord}_z(f) \cdot z = \frac{1}{2\pi i} \int_{\partial(\alpha+\Pi)} \frac{f'}{f} z dz.$$

We select α such that f does not vanish on the boundary. For integrations, consider

$$\int_{\alpha+w_1}^{\alpha+w_1+w_2} \frac{f'}{f} z dz = \int_{\alpha}^{\alpha+w_2} \frac{f'}{f}(t + w_1)dt = \int_{\alpha}^{\alpha+w_2} \frac{f'}{f} t dt + w_1 \int_{\alpha}^{\alpha+w_2} \frac{f'}{f} t dt.$$

Similarly,

$$\int_{\alpha+w_2}^{\alpha+w_1+w_2} \frac{f'}{f} z dz = \int_{\alpha}^{\alpha+w_1} \frac{f'}{f} t dt + w_2 \int_{\alpha}^{\alpha+w_1} \frac{f'}{f} t dt.$$

This implies

$$\sum_{z\in\alpha+\Pi} \mathrm{ord}_z(f) \cdot z = \frac{1}{2\pi i} \int_{\partial(\alpha+\Pi)} \frac{f'}{f} z dz = w_1 \int_{\alpha}^{\alpha+w_2} \frac{f'}{f} t dt + w_2 \int_{\alpha}^{\alpha+w_1} \frac{f'}{f} t dt.$$

On the other hand,

$$\frac{1}{2\pi i} \int_{\alpha}^{\alpha+w_1} \frac{f'}{f} dz = \frac{1}{2\pi i} \int_{f\gamma} \frac{dz}{z} \in \mathbb{Z}$$

and similarly

$$\frac{1}{2\pi i} \int_{\alpha}^{\alpha+w_2} \frac{f'}{f} dz \in \mathbb{Z}.$$

Consequently,

$$\sum_{z\in\alpha+\Pi} \mathrm{ord}_z(f) \cdot z \in L.$$

□

Remark 1.5. Proposition 1.4 will be interpreted as a weighted average of zeros and poles of a given elliptic function.

2. Example of elliptic function: Weierstrass p-function

Define Weierstrass function

$$\wp(z) = \frac{1}{z^2} + \sum_{w \in L \setminus \{0\}} (\frac{1}{(w-z)^2} - \frac{1}{w^2}).$$

It is evidently even. To see it is elliptic, consider

$$\wp'(z) = \sum_{w \in L} \frac{-2}{(z-w)^3},$$

which is elliptic. That is Therefore,

$$\wp(z + w_i) - \wp(z) = constant, i = 1, 2.$$

This constant is 0 since $\wp$ is even. To sum up, $\wp \in \mathcal{E}_L$.

Remark 2.1.
This function is convergent; we will leave the details to readers. See also [2][Chapter 9].

Remark 2.2. Given $R > 0$, for $|z| < R$, consider

$$\wp(z) = \frac{1}{z^2} + \sum_{0<|w|<2R} + \sum_{|w|>2R} \left(\frac{1}{(z-w)^2} - \frac{1}{w^2} \right).$$

Then the second term is holomorphic and the first term exhibits poles of second degree at those lattice points in the disc $B(0, R)$. Therefore, $\wp$ has poles at L.

Next, we deduce an essential property between $\wp$ and $\wp'$.
To this end, let's revoke the identity

$$\frac{1}{(1-x)^2} = \sum_{k=0}^{\infty} (k+1) x^k.$$

Thus, we have

$$\begin{aligned}
\wp(z) &= \frac{1}{z^2} + \sum_{w \in L^*} \left(\frac{1}{w^2} \left(\sum_{k=0}^{\infty} (k+1) \left(\frac{z}{w} \right)^k - \frac{1}{w^2} \right) \right) \\
&= \frac{1}{z^2} + \sum_{w \in L^*} \sum_{k=1}^{\infty} (k+1) \left(\frac{z^k}{w^{k+2}} \right) \\
&= \frac{1}{z^2} + \sum_{k=1}^{\infty} G_{k+2} (k+1) z^k \\
&= \frac{1}{z^2} + \sum_{k=1}^{\infty} G_{2k+2} (2k+1) z^{2k}.
\end{aligned}$$

Here, we denote by $G_k \coloneqq \sum_{w \in L \setminus \{0\}} \frac{1}{w^k}$.

Proposition 2.3.

$$(\wp')^2 = 4\wp^3 - g_2 \wp - g_3,$$

where $g_2 = 40G_4$ *and* $g_3 = 140G_6$.

Proof. Notice

$$\begin{aligned}
\wp &= \frac{1}{z^2} + 3G_4 z^2 + 5G_6 z^4 + \dots, \\
\wp^2 &= \frac{1}{z^4} + 6G_4 + 10G_6 z^2 + \dots, \\
\wp^3 &= \frac{1}{z^6} + \frac{9G_4}{z^2} + 15G_6 + \dots, \\
\wp' &= -\frac{2}{z^3} + 6G_4 z + 20G_6 z^3, \\
(\wp')^2 &= \frac{4}{z^6} - \frac{24G_4}{z^2} - 80G_6 + \dots.
\end{aligned}$$

Therefore, we find that

$$\begin{aligned}
(\wp')^2 - 4\wp^3 + 60G_4\wp &= \left(\frac{4-4}{z^6}\right) + \left(\frac{-24G_4 - 36G_4 + 60G_4}{z^2}\right) + (-80G_6 - 60G_6) + \dots \\
&= -140G_6 + \dots.
\end{aligned}$$

Since $(\wp')^2 - 4\wp^3 + 60G_4\wp$ is a holomorphic elliptic function, thus constant, it follows that

$$(\wp')^2 - 4\wp^3 + 60G_4\wp = -140G_6 = -g_3$$

by Liouville's theorem. □

3. Rational Parametrization and Group Structure

One can use the Weierstrass p-function to parametrize elliptic curves as follows:

$$\varphi : \mathbb{C}/L \to X(g_2(L), g_3(L)), \quad w \mapsto \begin{cases} [\wp(w) : \wp'(w) : 1], & w \notin L, \\ [0:1:0], & w \in L, \end{cases}$$

where $X(a,b)$ denotes the zero set of $F(X,Y,Z) = ZY^2 - 4X^3 + aXZ^2 + bZ^3$ in $\mathbb{P}^2$.

Fact: If $a^3 - 27b^2 \neq 0$, then there exists a lattice L with $g_2(L) = a$ and $g_3(L) = b$.

Proposition 3.1. *The map φ is an isomorphism between $\mathbb{C}/L$ and $X(g_2(L), g_3(L))$.*

Proof details can be adapted as needed. Here we focus on the lemmas and propositions formatting.

Lemma 3.2. *If $\wp(w_1) = \wp(w_2)$, then either $w_1 - w_2$ or $w_1 + w_2$ belongs to L.*

Proof. Suppose $w_1 - w_2 \notin L$. Consider the function $f(w) = \wp(w) - \wp(w_1) \in \mathcal{E}_L$. It has poles of second order at L. Therefore $f(w)$ has exactly two simple zeros at w_1, w_2 by Lemma 1.3. We can find $\alpha \in \mathbb{C}$ such that $\alpha + \Pi$ containing 0 and $\partial(\alpha + \Pi) \cap P(f) =$. We may choose $w_1', w_2' \in \alpha + \Pi$ where $w_1' - w_1, w_2' - w_2 \in L$. Revoking Lemma 1.4, we have that

$$\sum_{z \in \alpha + \Pi} ord_z(f) \cdot z = w_1' + w_2' - 2 \cdot 0 \in L$$

if $f \in \mathcal{E}_L$. That is $w_1 + w_2 \in L$. □

Lemma 3.3. *Let X be a compact Riemann surface and $f \in \mathcal{M}(X)$, the meromorphic function on X. If f is non-constant, then it is surjective.*

Proof. Confer [1][Theorem 2.7]. □

Proof of Proposition 3.1. **Injectivity:** Suppose $w_1, w_2 \in \mathbb{C}/L$ with $\varphi(w_1) = \varphi(w_2)$. If $\wp(w_1) = \wp(w_2)$, then either $w_1 - w_2$ or $w_1 + w_2 \in L$ by Lemma 3.2. Assume $w_1 + w_2 \in L$. Since $\wp'$ is odd, $\wp'(w_1) = -\wp'(-w_1) = -\wp'(w_2)$. If this term is non-zero, it contradicts $\varphi(w_1) = \varphi(w_2)$. If this term is zero, then $2w_1 \in L$, which implies $w_1 = w_2$.

Surjectivity: By Lemma 3.3, non-constant $\wp$ is surjective onto $\mathbb{P}^1$. Thus for any point $[x : y : 1] \in X(a, b) \cap U_2$, there exists a $w \in \mathbb{C}/L$ such that $\wp(w) = x$. By the formulas $(\wp')^2 = 4\wp^3 - a\wp - b$ and $y^2 = 4x^3 - ax - b$, it follows that either $(\wp(w), \wp'(w))$ or $(\wp(-w), \wp'(-w))$ equals (x, y), ensuring surjectivity.

Holomorphicity: For $w \notin L$, choose any neighborhood W excluding the origin with chart α, and let U_2 be the standard coordinate neighborhood with chart π_2. The composition $\pi_2 \circ \varphi \circ \alpha^{-1} = (\wp(\cdot), \wp'(\cdot))$ is holomorphic. Similarly, for $w = 0$, choose a neighborhood W without zeros of $\wp'$, with chart β and using neighborhood U_0, we find

$$\pi_0 \circ \varphi \circ \beta^{-1} = \left(\frac{\wp(\cdot)}{\wp'(\cdot)}, \frac{1}{\wp'(\cdot)}\right)$$

is also holomorphic. This confirms the holomorphicity of φ. □

For the moment, we are trying to transfer the Abelian group structure from $(\mathbb{C}/L, +)$ to $X(a, b)$.

Lemma 3.4. *If $u, v, w \in \mathbb{C}$ with $u + v + w \in L$ and $\wp(u) \neq \wp(v)$, then*

$$\det \begin{pmatrix} 1 & \wp(u) & \wp'(u) \\ 1 & \wp(v) & \wp'(v) \\ 1 & \wp(w) & \wp'(w) \end{pmatrix} = 0.$$

Proof. Without loss of generality, We may assume $u, v, w \in L$.

If $\wp(u) \neq \wp(v)$, then there exists a unique pair $(a, b) \in \mathbb{C}^2$ such that

$$\begin{cases} b + a\wp'(u) = \wp(u), \\ b + a\wp'(v) = \wp(v). \end{cases} \tag{3.1}$$

On one hand, u, v are two distinct roots of the elliptic function $f = b + a\wp'(u) - \wp(u)$. This function, having poles of multiplicity three at L, must also have an extra simple zero.

Let us find $\alpha \in \mathbb{C}$ such that $\alpha + \Pi$ contains 0 and $\partial(\alpha + \Pi) \cap P(f) = \emptyset$. Choose $u', v' \in \alpha + \Pi$ where $u' - u, v' - v \in L$. Referring to the lemma above, we have $\sum_{z \in \alpha + \Pi} \operatorname{ord}_z(f) \cdot z = 0$ if $f \in \mathcal{E}_L$. If w' is the third zero of f, then $u' + v' + w' + 3 \cdot 0 = 0$. Hence, $w' = -(u' + v')$ and thus $w' - w \in L$. This results in the system

$$\begin{cases} b + a\wp'(u) = \wp(u), \\ b + a\wp'(v) = \wp(v), \\ b + a\wp'(-(u + v)) = \wp(-(u + v)), \end{cases} \tag{3.2}$$

being linearly dependent, or equivalently,

$$\det \begin{pmatrix} 1 & \wp(u) & \wp'(u) \\ 1 & \wp(v) & \wp'(v) \\ 1 & \wp(u + v) & -\wp'(u + v) \end{pmatrix} = 0.$$

□

To transition the group structure from $\mathbb{C}/L$ to $X(a,b)$, we define the operation $p+q = \varphi(\varphi^{-1}(p)+\varphi^{-1}(q))$ with the identity element $o := \varphi^{-1}(0) = [0:1:0]$. Notice, the inverse of $p = [\wp(w), \wp'(w) : 1]$ is given by $-p = [\wp(-w), \wp'(-w) : 1] = [\wp(w), -\wp'(w) : 1]$.

Remark 3.5. We reinterpret the lemma in a more geometric manner: if $p+q+r = o$, then they are collinear. This confirms the group structure on $X(a,b)$ as dispicted in Figure 3.1.

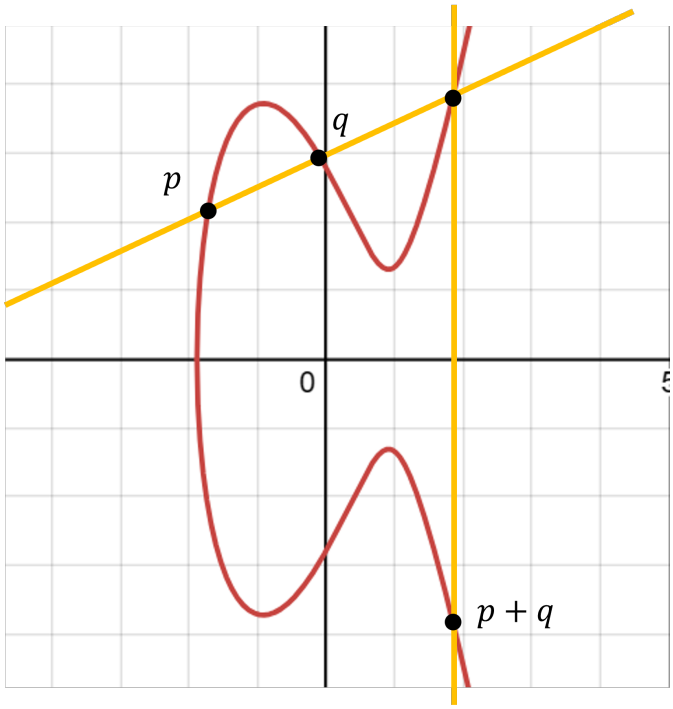


FIGURE 3.1. Group law on elliptic curve

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